

Experiment No- 4

Aim:

To perform OC & SC test on alternator.

Apparatus Required:

S no	Instruments Required	Rating	Quantity
1	DC Compound Motor	220 V 16 KW 85 A 750 - 2250 rpm	1
2	3 ϕ sine wave Alternator	400 V 15 KVA 1500 rpm	1
3	Ammeter	10/20 Amp MI, MC	1 each
4	Voltmeter	400 V MI, MC	1 each
5	Tachometer	0-2500 rpm	1

Theory:

The open-circuit test or the no-load test, is performed by driving the generator at its rated speed while the armature winding is left open. The field current is varied in suitable steps and the corresponding values of the open-circuit voltage varied in suitable steps and corresponding values of the open-circuit voltage between any two pairs of terminals of the armature windings are recorded.

The OCC follows a straight-line relation as long as the magnetic circuit of the synchronous generator does not saturate. In the linear region, most of the applied MMF is consumed by the air gap; the straight line is appropriately called the air-gap line. As the saturation sets in, the OCC starts deviating from the air-gap line.

The short-circuit test provides information about the current capabilities of a synchronous generator. It is performed by driving the generator at its rated speed when the terminals of the armature winding are shorted. An ammeter is placed in series with one of the three short lines. The field current is gradually increased and the corresponding value of the armature current is recorded. When the per-phase short-circuits current is plotted as a function of the field current, the graph is called the short circuit characteristic of a generator.

Voltage Regulation:

The voltage regulation of an alternator is defined as " the rise in voltage from full-load to no-load and (field excitation and speed remaining the same) divided by the rated terminal voltage".

$$\text{Percent regulation} = (E_o - V)/V * 100$$

Where E_o and V are respectively the no load emf and full load voltage.

Formulas:

Synchronous impedance, $Z_s = OA/OB$ = for field current I_{sc} [plot from graph $OA/OB = V_o/I_{sc}$]

I_{sc1} is selected over the linear part of OCC, generally, it corresponds to rated armature current.

Synchronous reactance, $X_s = \sqrt{(Z_s^2 - R_a^2)}$, where R_a = Armature resistance of alternator (per phase)

Calculate the excitation emf E_o and voltage regulation for full-load and

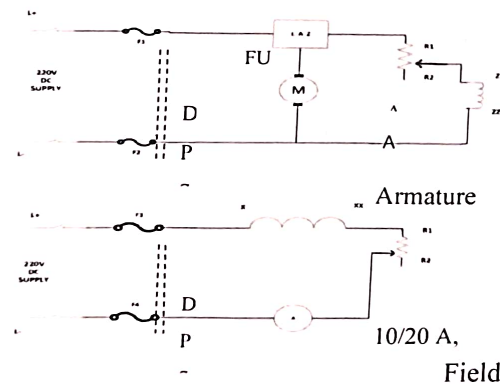
1. 0.8 lagging p.f.
2. UPF
3. 0.8 leading p.f.

$E_o = [(V \cos \phi + I_a R_a)^2 \mp (V \sin \phi + I_a X_s)^2]^{1/2}$ (+ sign is for lagging pf load, - sign is for leading pf load)
where V = rated terminal voltage per phase of alternator

$$\% \text{Regulation} = (E_o - V) / V * 100$$

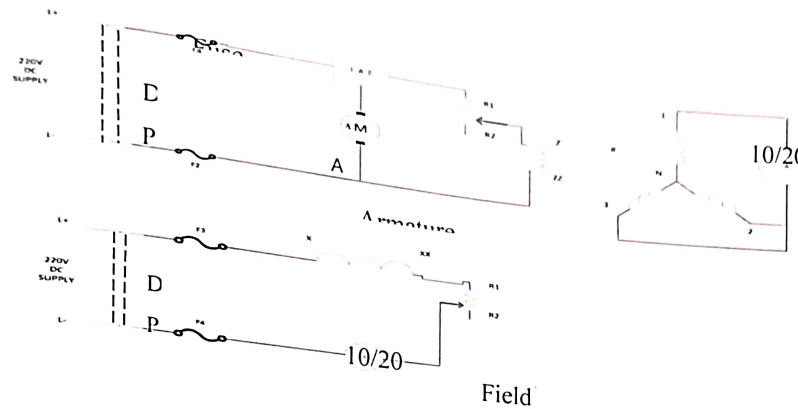
Circuit Diagram:

[A] Open Circuit Test



4
0
0

[B] Short Circuit Test



Procedure:

[A] Open Circuit Test

- Connect the circuit as shown in the figure.
- Set the potential divider to zero output position and motor field rheostat to the minimum value.
- Switch on the DC supply and start the motor.
- Adjust motor speed to synchronous value by motor field rheostat and note the meter readings.
- Increase the field excitation of the alternator and note the corresponding readings.
- Repeat step 5 till 10% above the rated terminal voltage of the alternator.
- Maintain constant rotor speed for all readings.

[B] Short Circuit Test

- Connect the circuit as shown in the figure.
- Star the motor with its field rheostat at minimum resistance position and the potential divider set to zero output.
- Adjust the motor speed to synchronous value.
- Increase the alternator field excitation and note ammeter readings.
- Repeat step 4 for different values of excitations (field current). Take readings up to the rated armature current. Maintain constant speed for all readings.
- Measure the value of armature resistance per phase R_a by multimeter or by the ammeter-voltmeter method.
- Plot the characteristics and find the synchronous impedance.

Observations:**[A] Open Circuit Test**

Sno	Field current I_f (in Amp)	Line Voltage supplied V_L (in V)	Voltage per phase = $V_L/\sqrt{3}$ (in V)
1			
2			
3			
4			
5			
6			

[B] Short Circuit Test

Sno	Field Current I_f (in Amp)	Short Circuit Current I_{sc} (in Amp)
1		
2		
3		
4		
5		
6		

Alternator armature resistance per phase R_a = ----- Ω Rotor speed = ----- RPM

Graph:

Plot the readings to draw following graphs. Use the same graph paper for both curves.

1. I_f versus V_o (from OC test) Plot Graph on Next Page
2. I_f versus I_{sc} (from SC test) Plot Graph on Next Page

Calculations:

Result:

Regulation of alternator at full load is found to be,

At unity pf = -----

At 0.8 lagging = -----

At 0.8 leading = -----

Synchronous Impedance varies for different values of excitation.

Precautions:

1. All connections should be perfectly tight and no loose wire should lie on the work table.
2. Before switching ON the dc supply, ensure that the starter's moving arm is at its maximum resistance position.
3. Do not switch on the supply, until and unless the connections are checked by the teacher
4. Avoid errors due to parallax while reading the meters.
5. Hold the tachometer with both hands steady and in line with the motor shaft so that it reads correctly.
6. Ensure that the winding currents do not exceed their rated values.